Fourteen-target theorem audit: a Lean increment

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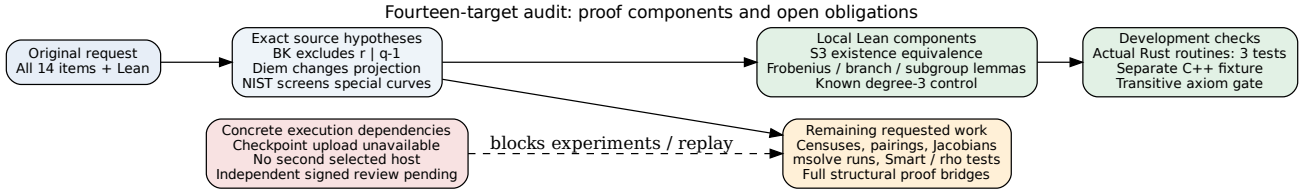
The full short-Weierstrass S3 zero-sum equivalence, including repeated coordinates, now compiles using Mathlib's actual elliptic point group. The theorem assumes an algebraically closed field with 2 nonzero, short coefficients and nonzero discriminant. This is the central general S3 converse; it is not a binary formula or a resultant-degree theorem. Signed PCC standing is pending. [pcc:unverified]



The project contains 33 new declarations and reuses two unchanged historical modules. The repository axiom gate passes all 68 audited declarations (33 new, 35 reused), with zero forbidden axioms and zero native exceptions. Local builds, production Rust regression tests and axiom outputs are retained in evidence/. The proof map names every statement and its limits; the full README retains all fourteen requirements. [pcc:unverified]

Targets	Component supplied	Still required
1	General S3 existence equivalence	binary case, recursion/degrees, native exhaustive checks
2–3	Triangular Frobenius core; exact order from trace-zero cardinality	actual torsion/Frobenius and supersingular/Hasse bridges; extension census
4–5	Cyclic pairing restriction; stable branch-locus obstruction	actual pairings, ramification geometry, factor-base lower bound
6–7	Asymptotic source review; abstract subgroup confinement	Jacobian definitions, counts and counterexample family
8–9	Nonreduced multiplicity control; exact hypothesis review	general multihomogeneous bound; msolve system experiments
10–11	One-step collision mass; walk and NIST source review	mixing/collision analysis, Smart attack and rho measurements
12–13	Pinned msolve source audit; existing cross-language evidence	implementation refinement and second-machine replay
14	Known degree-3 proof and actual census-routine controls	independent signed validation

Every entry is a scoped producer assertion, not a signed scientific promotion. The original request remains incomplete. [pcc:unverified]

The fixed negative control is F={x0^2+x0, x1^2+x1, x0*x1+x0+x1} over F2. The existing criterion rejects degree two; explicit degree-three rows contain the Gröbner basis {x0,x1}. The actual Rust functions

return low dimensions 0 and2, ranks3 and9; all three fixed development tests pass. An independent C++ fixture agrees on this same host. [pcc:unverified]

Source corrections matter: the actual BK node excludes r dividing q-1, so the earlier F7 examples do not refute it. Diem Condition2.7 excludes the base-defined x projection by its Frobenius-stable branch locus; Diem constructs another projection. NIST AppendixC.1 excludes anomalous curves and requires large embedding degree. See sources.json for exact hashes and read sections. [pcc:unverified]

No paid cloud jobs or new research censuses were launched. The complete resume document could not be uploaded because the required checkpoint token and AWS profile are absent. No second independent execution environment is selected. These dependencies block experiments/replay, not the completed local proof and development checks. [pcc:unverified]

Reproduction: `bash check.sh`; then the repository `tools/lean/proof-gate.sh research/theorem_audit_20261108/lean`. Lean4.35.0 and Mathlib56a005855ea17e0b0958f73ec801ae140b0a1679 are pinned. The accepted axiom set is propext, Classical.choice and Quot.sound; no native evaluation or research axiom is permitted. [pcc:unverified]